

MATRIX ACADEMY

- 5/11, Rajender Nagar, Sec-5, Opp. DAV Public School, Sahibabad, Ghaziabad, U.P.
- S-14, First Floor, Opp. S.M. World, Shalimar Garden-II, Sahibabad, Ghaziabad, U.P.

Name Shivansh Sharma Class & Batch Xth 3-4pm Subject Maths Date 23/08/26

Section-A

Q1.

$$D = b^2 - 4ac$$

for real & distinct roots, $D > 0$
 $\Rightarrow b^2 - 4ac > 0$

$$\Rightarrow (-8)^2 - 4(1)(k) > 0$$

$$64 - 4k > 0$$

$$+4k > +64$$

$$k > 16$$

Ans $\rightarrow$ D) $k > 16$

3

$$\begin{array}{r} 2 \overline{) 31} \\ 2 \overline{) 16} \\ \underline{2} \\ 2 \\ \underline{2} \\ 2 \\ \underline{2} \\ 0 \end{array}$$

$$750^\circ$$

Q2

$$D = b^2 - 4ac$$

for rational roots, $D > 0$

$$b^2 - 4ac > 0$$

$$(k)^2 - 4(2)(-4) > 0$$

$$k^2 + 32 > 0$$

$$k^2 > -32$$

$$k > \pm \sqrt{32}$$

(C) +2

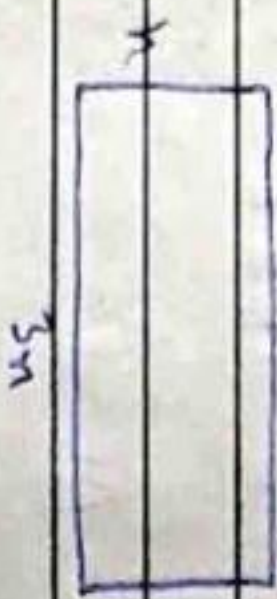
Q3. (c) $BC \parallel XY$ ✓

Q4. (B) 27 put ✓

Q5. (A) Both A & R are true and R is correct explanation of A. ✓

Section-B

Q6.



Let the breadth rectangular field is m
then it's length $\rightarrow 3m$
area = ~~147~~ $147 m^2$

$$m(3m) = 147$$

$$3m^2 - 147 = 0$$

$$3(m^2 - 49) = 0$$

$$m^2 - 49 = 0$$

$$(m)^2 - (7)^2 = 0$$

$$(m-7)(m+7) = 0$$

$\Rightarrow$

$$m = 7m$$

$$m = -7m$$

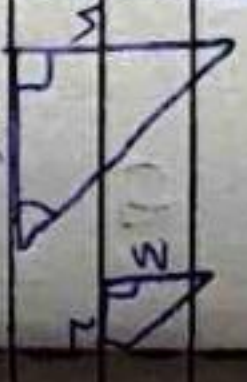
rejected, as breadth couldn't be negative.

rough

$$\begin{array}{r} 2.6 \\ 3.9 \overline{) 8.5} \\ \underline{7.8} \\ 0.7 \end{array}$$

$$\begin{array}{r} 2.6 \\ 3.9 \overline{) 10.14} \\ \underline{7.8} \\ 2.34 \\ \underline{2.34} \\ 0 \end{array}$$

$$\begin{array}{r} 2.6 \\ 3.9 \overline{) 10.14} \\ \underline{7.8} \\ 2.34 \\ \underline{2.34} \\ 0 \end{array}$$



$$\frac{3}{18} = \frac{4}{18}$$

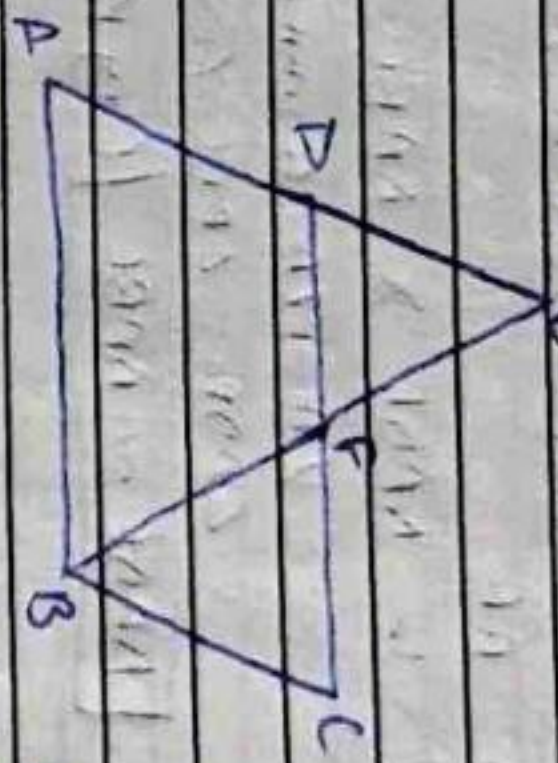
$$m = \frac{3 \times 18}{2}$$

$$m = 27$$

Q7.

$\Rightarrow$ breadth $\Rightarrow 7m$

$\therefore$ length $\Rightarrow 3(7) = 21m$ Ans.



given - $ABCD$ is a $\parallel gm$, AD is produced to F , BF intersects CD at F .
to prove - $\triangle ABE \sim \triangle CFB$.

proof -

Since $ABCD$ is a $\parallel gm$,
so, $AD \parallel CB$

$\therefore$ AB and is produced to F

$\therefore$ AE is also $\parallel$ to CB

now, in $\triangle ABE \sim \triangle CFB$ -

$\angle AEB = \angle CFB$ (alt. $\angle$ s, $AE \parallel CB$ & BE transversal)

$\angle EAB = \angle BCF$ ($\therefore$ opp. $\angle$ s of $\parallel gm$ are equal)

$\Rightarrow$ $\triangle ABE \sim \triangle CFB$ (by AA)

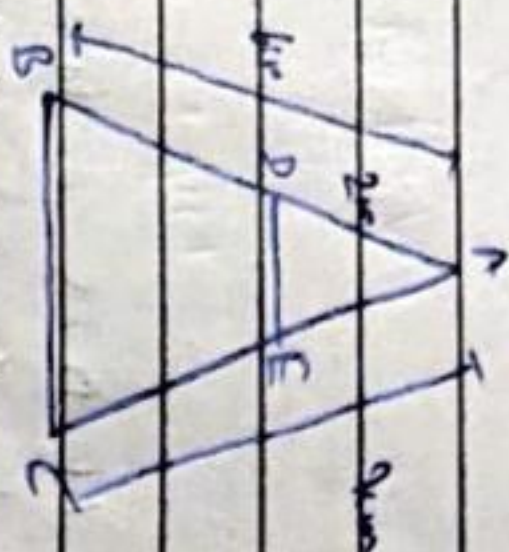
MATRIX ACADEMY

• 5/11, Rajender Nagar, Sec-5, Opp. DAV Public School, Sahibabad, Ghaziabad, U.P.
• S-14, First Floor, Opp. S.M. World, Shalimar Garden-II, Sahibabad, Ghaziabad, U.P.
Name Shikharish Sharma Class & Batch XIII 34P13 Subject Maths

Date 23/06/22

Ravet Sir

Q8.



given in $\triangle ABC$, $DE \parallel BC$.

$AD = 2\text{cm}$, $AB = 6\text{cm}$ & $AC = 9\text{cm}$

to find $\rightarrow$ AE

solution. in $\triangle ADE$ & $\triangle ABC$ -

$\angle A = \angle A$ (common)

$\angle ADE = \angle ABC$ (corresponding $\angle$'s)

$\Rightarrow \triangle ADE \sim \triangle ABC$ (by AA)

$\frac{AD}{AB} = \frac{AE}{AC}$ (cpct)

$\Rightarrow \frac{2}{6} = \frac{AE}{9}$

$30 = 6AE$

$\frac{1}{2} = \frac{AE}{9}$

$AE = 3\text{cm}$

Section-1)

Q9.

S.P. of article $\rightarrow$ $\text{Rs } 75$ distance $= 300\text{km}$

Let the C.P. be $\text{Rs } x$ Let original speed $= x \text{ km/hr}$

Profit $= \frac{x(100)}{100} + y$ $t = \frac{300}{x}$

$75 - x = \frac{x^2}{100}$ now, If speed $= x + 5 \text{ km/hr}$

$\Rightarrow \frac{x^2}{100} + x + 5 - 75 = 0$ then $t' = \frac{300}{x+5}$

$\Rightarrow \frac{x^2}{100} + x - 70 = 0$ $t - t' = 2$

$\Rightarrow \frac{x^2}{100} + x - 70 = 0$ $\Rightarrow \frac{300 - 300}{x+5} = 2$

$\Rightarrow x^2 + 20x - 7000 = 0$ $\Rightarrow \frac{300(x+5) - 300(x)}{x+5} = 2$

$\Rightarrow x^2 + 20x - 7000 = 0$ $\Rightarrow \frac{300(x+5) - 300(x)}{x+5} = 2$

$\Rightarrow x^2 + 20x - 7000 = 0$ $\Rightarrow \frac{300(x+5) - 300(x)}{x+5} = 2$

$\Rightarrow x^2 + 20x - 7000 = 0$ $\Rightarrow \frac{300(x+5) - 300(x)}{x+5} = 2$

$\Rightarrow x^2 + 20x - 7000 = 0$ $\Rightarrow \frac{300(x+5) - 300(x)}{x+5} = 2$

$\Rightarrow x^2 + 20x - 7000 = 0$ $\Rightarrow \frac{300(x+5) - 300(x)}{x+5} = 2$

$\Rightarrow x^2 + 20x - 7000 = 0$ $\Rightarrow \frac{300(x+5) - 300(x)}{x+5} = 2$

$\Rightarrow x^2 + 20x - 7000 = 0$ $\Rightarrow \frac{300(x+5) - 300(x)}{x+5} = 2$

$\Rightarrow x^2 + 20x - 7000 = 0$ $\Rightarrow \frac{300(x+5) - 300(x)}{x+5} = 2$

57500

51300

5260

552

226

213

31

5750

5130

550

56

512

512

512

512

512

512

512

512

512

512

512

512

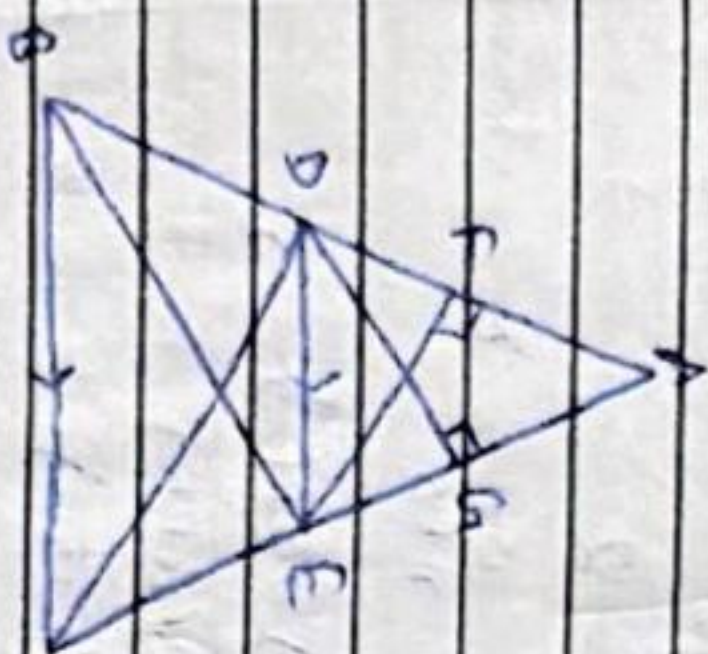
$$\Rightarrow n(n+30) - 25(n+30) = 0$$

$$\Rightarrow (n+30)(n-25) = 0$$

$\Rightarrow \boxed{n = -30}$ or $\boxed{n = 25}$
 Δ rejected as speed cannot be negative

$\therefore$ the usual speed of train was 25 km/hr Ans.

Q11.



given: In $\triangle ABC$, $DE \parallel BC$
 to prove: $\frac{AD}{BD} = \frac{AE}{EC}$

construction: Join BF & CD

proof: Draw $EF \parallel AD$ & $DG \perp AE$
 area of $\triangle ADE = \frac{1}{2} \times AD \times EF$
 area of $\triangle BDE = \frac{1}{2} \times BD \times EF$

$$\Rightarrow \frac{\text{area } \triangle ADE}{\text{area } \triangle BDE} = \frac{AD}{BD} \quad \text{--- (1)}$$

now, $\frac{\text{area } \triangle ADE}{\text{area } \triangle CEB} = \frac{\frac{1}{2} \times AE \times EF}{\frac{1}{2} \times EC \times EF} = \frac{AE}{EC} \quad \text{--- (2)}$

now, we know that two triangles with same base & equal height have the same area.

$$\Rightarrow \underline{\text{area } \triangle CEB = \text{area } \triangle BDE}$$

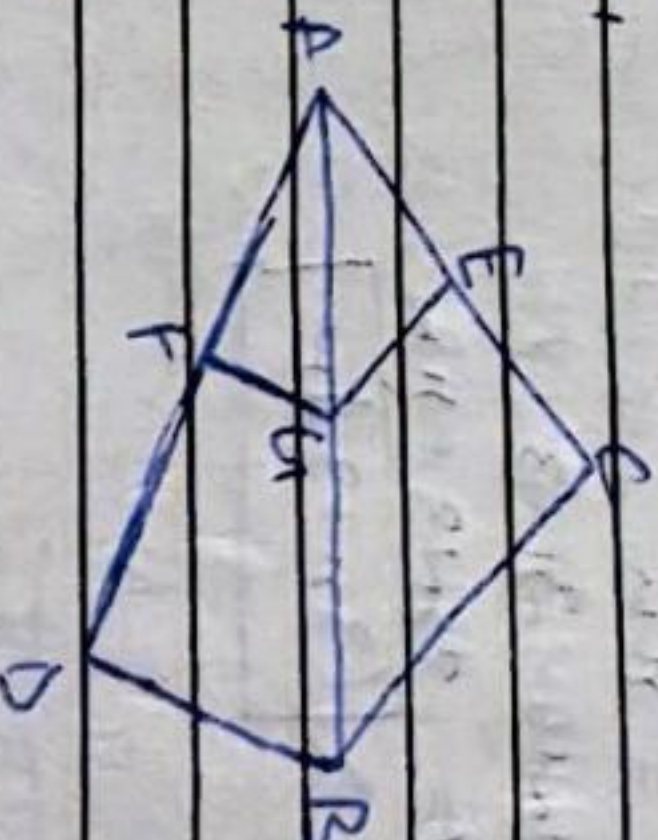
$$\Rightarrow \therefore \frac{\text{area } \triangle ADE}{\text{area } \triangle BDE} = \frac{\text{area } \triangle ADE}{\text{area } \triangle CEB}$$

from (1) & (2)

$$\boxed{\frac{AD}{BD} = \frac{AE}{EC}}$$

hence, proved.

now,



given: $EF \parallel CB$ & $EF \parallel DB$
 to prove: $\frac{AE}{EC} = \frac{AF}{FB}$

proof: In $\triangle ABC$, $EF \parallel CB$, so by BPT-

$$\frac{AE}{EC} = \frac{AF}{FB} \quad \text{--- (1)}$$

MATRIX ACADEMY

• 5/11, Rajender Nagar, Sec-5, Opp. DAV Public School, Sahibabad, Ghaziabad, U.P.
• S-14, First Floor, Opp. S.M. World, Shalimar Garden-II, Sahibabad, Ghaziabad, U.P.
Name Shreshth Sharma Class & Batch XII-3-4 Subject Maths

Date 23/08/24

Page No. 57

In $\triangle ABD$, $\triangle HGB$ So by BPT -

$$\Rightarrow \frac{AF}{FD} = \frac{AG}{GB} \quad \text{--- (1)}$$

from (1) & (2)

$$\Rightarrow \frac{AE}{EC} = \frac{AF}{FD}$$

hence proved.

Section-E

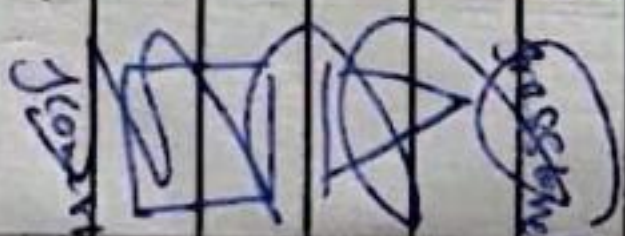
Q1 (i)

length of square (Glowabed) = n m

then the length of glowabed = $2n+3$

~~then~~ $\Rightarrow$ total length of glow = $2n+3+n$

$$\Rightarrow (3n+3) \text{ m}$$



(ii)

perimeter of field $\Rightarrow 2(l+b)$

$$\Rightarrow 2[4n+3] + 2[3n+3+n]$$

$$\Rightarrow 2[4n+3]$$

$$\Rightarrow [8n+6 \text{ m}]$$

(iii)

area of total field = 1260 m^2

$$\Rightarrow l \times b = 1260 \text{ m}^2$$

$$\Rightarrow (3n+3)(n) = 1260$$

$$\Rightarrow 3n^2+3n = 1260$$

$$\Rightarrow 3n^2+3n-1260=0$$

$$\Rightarrow 3(n^2+n-420)=0$$

$$\Rightarrow n^2+n-420=0$$

$$\Rightarrow n^2+21n-20n-420=0$$

$$\Rightarrow n(n+21)-20(n+21)=0$$

$$\Rightarrow (n+21)(n-20)=0$$

$$\Rightarrow [n=-21] \text{ or } [n=20 \text{ m}]$$

rejected

$$[n=20 \text{ m}]$$

$$\begin{array}{r} 2 \overline{) 420} \\ \underline{2} \\ 2 \\ \underline{2} \\ 0 \end{array}$$

Section C

Q9

$$S.C. = ₹75$$

$$Profit = 75 - n$$

$$0.150, Profit = n \cdot 0.150 = \frac{n(150)}{1000}$$

$$\Rightarrow \frac{n(150)}{1000} - 75 - n$$

$$\Rightarrow \frac{n^2}{100} + n - 75 = 0$$

$$\Rightarrow n^2 + 100n - 7500 = 0$$

$$n = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\Rightarrow n = \frac{-100 \pm \sqrt{(100)^2 - 4(-7500)(1)}}{2(1)}$$

$$2(1)$$

$$\Rightarrow n = \frac{-100 \pm \sqrt{10000 + 30000}}{2}$$

$$2$$

$$\frac{75}{1000}$$

$$\frac{2000}{1000}$$

$$n = \frac{-100 \pm \sqrt{40000}}{2}$$

$$n = \frac{-100 \pm 200}{2}$$

$$n = \frac{-100 - 200}{2} \quad \& \quad n = \frac{-100 + 200}{2}$$

$$n = \frac{-300}{2} \quad n = \frac{100}{2}$$

$$n = -150$$

$$n = 50$$

Rejected

$\therefore$ the cost price of the article is ₹50.